

and when he is paying fixed, he wants to pay the lowest fixed rate he can. Of course, swap dealers have their own incentives of earning bid/offer when they make markets in swaps and thus will want to pay at the lowest rate possible to clients and receive at the highest rates possible from clients. The forces of competition between dealers govern the bid/offer spread in the market.

Table 1.1 is an example of a composite page of swap spreads and swap rates, showing the market for swaps of various maturities. Such composite pages are made available by a variety of companies, including Tradeweb Markets LLC and Thomson Reuters. The data presented in such a page are an amalgamation of market levels submitted by a number of dealers and provide an indication of where one could expect to execute various swaps. Market levels on the page are continuously updated throughout the trading day.

Each row of this composite page presents information on the market for swaps of a particular maturity. It is important to note that all of the information about swaps on this page assumes that the floating side of the swap is 3-month LIBOR paid quarterly Act/360. If the floating side of the swap is anything other than 3-month LIBOR paid quarterly Act/360, this could have significant implications on what the appropriate fixed rate in the swap is going to be, as we will see.

The group of numbers under the column labeled “Treasury Prices” gives information for the market in on-the-run Treasuries. The *bid price* in the market is the highest price at which a Treasury dealer is willing to buy Treasuries, and the *offer price* (also sometimes called *ask price*) is the lowest price at which a Treasury dealer is willing to sell Treasuries. For instance, the bid price for the 5-year on-the-run Treasury is $101-15\frac{3}{4}$, and the offer price is $101-16$.¹⁴ Note that the yield corresponding to the bid price is 1.438%, and the yield corresponding to the offer price is 1.436%.¹⁵ The next column labeled “Mid Yield” gives the mid-market yield on Treasuries, defined to be the arithmetic average of the bid and the ask yields. Thus, for example, in five years, the mid-market yield is $(1.438\% + 1.436\%)/2 = 1.437\%$.

The column labeled “Swap Spread” provides market information for swap spreads. The left-hand number is the *offer side spread*, the lowest spread that a dealer is willing to add to the Treasury yield to receive fixed, and the right-hand number is the *bid side spread*, the highest spread that a dealer is willing

¹⁴Treasury prices are quoted in dollars and 32nds of a dollar. Each 32nd may be further subdivided into 8ths. Thus a price of $101-15\frac{3}{4}$ represents a price of 101 and 15.75 32nds, or 101.4921875% of par. The price on the screen is the *quoted price*, which does not include accrued interest. The *dirty price*, or *full invoice price*, represents the full proceeds one pays for the bond, which includes accrued interest.

¹⁵Since Treasury prices and yields move inversely, the higher Treasury price of course has a lower corresponding yield.

Term	Treasury Prices	Treasury Yields	Mid Yield	Swap Spread	Semi Bond	Annual Money
2-year	100-06 $\frac{3}{4}$ -100-07	0.517-0.513	0.515	19.00/15.00	0.705-0.665	0.696-0.656
3-year	99-29 $\frac{1}{4}$ -99-29 $\frac{1}{2}$	0.779-0.776	0.778	23.00/19.00	1.008-0.968	0.996-0.956
4-year	—	—	1.107	22.25/18.25	1.329-1.289	1.315-1.276
5-year	101-15 $\frac{3}{4}$ -101-16	1.438-1.436	1.437	21.00/17.00	1.647-1.607	1.630-1.591
6-year	—	—	1.758	18.75/14.75	1.946-1.906	1.927-1.887
7-year	101-29-101-29 $\frac{1}{2}$	2.080-2.077	2.078	11.50/7.50	2.193-2.153	2.174-2.135
8-year	—	—	2.283	11.25/7.25	2.395-2.355	2.376-2.336
9-year	—	—	2.488	7.75/3.75	2.566-2.526	2.546-2.506
10-year	106-28-106-28 $\frac{1}{2}$	2.694-2.692	2.693	1.75/-2.25	2.710-2.670	2.690-2.650
12-year	—	—	2.693	25.50/21.50	2.948-2.908	2.928-2.888
15-year	—	—	3.003	19.50/15.50	3.198-3.158	3.178-3.138
20-year	—	—	3.314	10.25/6.25	3.416-3.376	3.397-3.357
25-year	—	—	3.625	-9.25/-13.25	3.533-3.493	3.513-3.473
30-year	107-21-107-21 $\frac{1}{2}$	3.936-3.935	3.935	-35.25/-39.25	3.583-3.543	3.563-3.523

Table 1.1: Sample Swap Composite Page. Used by Permission of Thomson Reuters and BrokerTec

to add to the Treasury yield to pay fixed.¹⁶ Thus the market for 5-year swap spreads is 21/17, meaning that the offer side is 21 bps, and the bid side is 17 bps. So the bid/offer spread in 5-year swaps at this moment in time is 21 bps – 17 bps = 4 bps wide, according to this page. There are times when the bid/offer spread is much tighter, reflecting a very liquid market, and times when it is wider, indicating a great deal of illiquidity in the market. Given the market depicted, if a dealer in swaps could simultaneously receive fixed in a 5-year swap at a fixed rate of $T_5 + 21$ and pay fixed in a 5-year swap at a fixed rate of $T_5 + 17$, the dealer could lock in a 4 bp *running* (i.e., an annuity over the life of the trade) profit. It is very seldom that things work out so smoothly in the swaps market, however, that a dealer has two trades in hand simultaneously that perfectly offset one another, allowing the dealer to lock in a riskless profit.

The next column of numbers labeled “Semi Bond” gives both the *offer side swap rate* (the lowest rate at which a dealer is willing to receive fixed) and the *bid side swap rate* (the highest rate at which a dealer is willing to pay fixed). These rates are quoted assuming the fixed side pays semiannually on a 30/360 basis. The bid side and offer side swap rates are computed using the bid side and offer side swap spreads, respectively, added to the mid-market yield of the relevant Treasury.¹⁷ The next column labeled “Annual Money” provides swap rates in which the fixed side is quoted annually Act/360.

The U.S. Treasury currently issues notes maturing in 2, 3, 5, and 7 years on a monthly auction schedule and 10-year notes and 30-year bonds on a quarterly schedule (with monthly reopenings of these latter two). Swap spreads for swaps of each of these benchmark maturities are quoted as a spread to the respective on-the-run Treasury yield. All other swap rates, with the exception of the 12-year swap rate, are computed as follows:

- Linear interpolation between the two adjacent on-the-run Treasury issues is used to compute an interpolated mid-market Treasury yield for that maturity.¹⁸

¹⁶It is commonplace when speaking about a swap to refer only to the fixed side. If it is stated that a counterparty is paying, that means the counterparty is paying fixed and receiving floating. If it is stated that a counterparty is receiving, that means that the counterparty is receiving fixed and paying floating. Thus, if someone shouts out across a trading floor and says that he wants to pay in a swap, you will never hear as a response, “Fixed or floating?” People would laugh at you if you did.

¹⁷This is a bit of a simplification. It is often more appropriate to think of the bid side swap rate as equal to the offered side yield on the Treasury plus the bid side swap spread, and the offer side swap rate as equal to the bid side Treasury yield plus the offer side swap spread. We will discuss this more shortly.

¹⁸Suppose T_L and T_H are Treasury yields for two adjacent on-the-run Treasuries of maturity L and H , respectively, and we wish to compute an M -year Treasury yield, T_M , where $L < M < H$, by linear interpolation. Then T_M is given by

- The appropriate swap spread for the given maturity is then added to that interpolated mid-market Treasury yield.¹⁹

Example

The 4-year swap spread (bid or offer) is added to the interpolated 4-year Treasury yield in order to come up with the market for 4-year swap rates. The interpolated 4-year Treasury yield is computed from the adjacent on-the-run 3-year and 5-year Treasury yields and is

$$0.778\% + \left(\frac{4-3}{5-3} \right) \times (1.437\% - 0.778\%) = 1.107\%,$$

as indicated in Table 1.1.²⁰ Thus, the market for 4-year swap rates is $1.107\% + 0.1825\% = 1.289\%$ on the bid side and $1.107\% + 0.2225\% = 1.329\%$ on the offer side. $\square$

Notice that 25-year swap spreads are actually negative in the composite page depicted in Table 1.1, in which case the swap rate for this maturity is actually lower than the interpolated Treasury yield for that maturity. Also notice 30-year swap spreads are negative on this page. The phenomenon of negative swap spreads was thought by many market participants to be an impossibility. In the USD swaps market, negative spreads had never occurred prior to 2008. But in the aftermath of the fall of Lehman Brothers during the financial crisis, 30-year swap spreads traded negative for the first time. This was a historic event, one that showed the stress and strain that the market was under and one that many people thought would soon correct itself when it first happened. But, as shown in Figure 1.3, it didn't; in fact, 30-year spreads (primarily) remained in negative territory for years thereafter. We shall return to this phenomenon in Chapter 9.

$$T_M = T_L + \left(\frac{M-L}{H-L} \right) \times (T_H - T_L).$$

¹⁹The convention in the market is that the 12-year swap spread is quoted as a spread to the yield of the 10-year Treasury. There is really no rhyme or reason to this. Market participants could just as easily quote the 12-year swap spread as a spread to the interpolated 12-year Treasury yield. In this case, the 12-year swap spread would have to be quoted lower (assuming the yield on the 30-year Treasury is above the yield on the 10-year Treasury) so as to leave the 12-year swap rate unaffected. It makes no difference as long as market participants know how they are being quoted.

²⁰Actually, the interpolated 4-year Treasury yield is 1.1075%, which has been rounded to 1.107%. We will employ such rounding frequently without further mention.

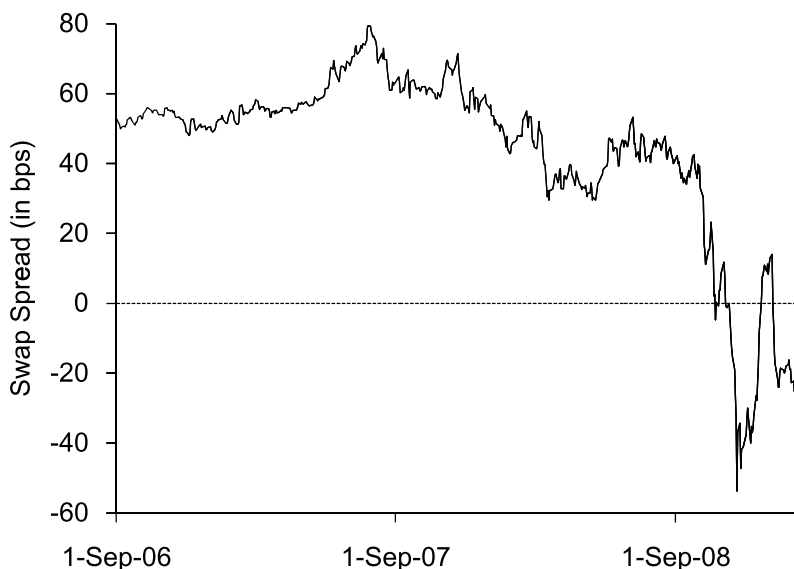


Figure 1.3: History of 30-Year Swap Spreads

Pay Frequency and Day Count

As mentioned previously, the standard convention in a swap is that the fixed side pays semiannually on a 30/360 basis, and the floating side pays 3-month LIBOR quarterly on an Act/360 basis, the entirety of which is often abbreviated “semi bond versus threes.”²¹ This is the standard unless the two counterparties agree otherwise. The terms of a swap can be arbitrarily specified as long as the two counterparties entering into the trade agree to them and the terms are priced accordingly. In this section we will consider potential changes to the fixed leg of a swap and the implications that this has on the fixed rate. It is very important to note that throughout this section we are considering changes only to the terms of the fixed leg; the floating leg is always assumed to pay 3-month LIBOR quarterly Act/360.²²

²¹Once again, “threes” is short for 3-month LIBOR paid quarterly Act/360. Other alternative LIBOR rates can be paid in a swap, such as “ones” (1-month LIBOR paid monthly Act/360) and “sixes” (6-month LIBOR paid semiannually Act/360). In addition, other floating rate indexes (such as Prime, T-Bills, Commercial Paper, etc.) are sometimes used as well.

²²In subsequent sections we will talk about changing the floating leg of the swap from 3-month LIBOR to another floating rate of interest and the impact that this has on the fixed coupon of the swap.

Although most swaps have a fixed side that pays semi bond, this need not be the case. Some swaps have a fixed side that pays monthly, or quarterly or annually. Some swaps, known as *zero coupon swaps*, only pay a fixed coupon one time at maturity. Not only can the payment frequency on the fixed side of a swap be changed from semiannual; it is also possible to accrue interest on the fixed side on a money basis or other bases, instead of the standard 30/360 basis.²³

A client might want to change the pay frequency on the fixed side so that the resultant fixed coupons in the swap match up with an asset or liability that the client is trying to hedge. For example, if a client has a fixed rate loan that pays monthly, he may want the fixed leg of a swap to pay monthly as well if he is going to want the cash flows from the fixed leg of the swap to match those of the loan. Any of these adjustments from the “plain vanilla” semi bond quote for a swap found on the composite page implies that the fixed coupon in the swap needs to be changed in order to make both counterparties indifferent to the change.

Suppose, for example, that a client is interested in executing a 5-year swap but wants the fixed coupon to be quarterly money instead of semi bond. What fixed rate should he expect in the swap? There is only one way to determine the correct answer to this question: use a swap model to price up the changes.²⁴ We will start to look at such models in Chapter 3. In the meantime, it is possible to come up with approximations that will get us in the ballpark of the correct answer.

Even though the only way to come up with the correct fixed rate given any change in payment frequency or day count is to use a pricing model, there is still a very good reason to be familiar with these approximations. An approximation can be compared with the result of a pricing model to provide a sanity check that something hasn’t been modeled incorrectly. If the price from a model and the result of an approximation computed on, say, a financial calculator are in close agreement, that should give comfort that the pricing was correct. If the two answers diverge significantly, that might suggest that it’s a good idea to check the model and make sure a mistake wasn’t made. So how close is close enough when it comes to comparing the result from a pricing model with the result from an approximation formula? That’s a personal question.²⁵ For many in the industry, an approximation

²³Other bases to accrue interest include Act/365 Fixed, for example, in which case the actual number of days in a period is divided by 365. But 30/360 and Act/360 are the two most popular in the United States.

²⁴Discrepancies in pricing may exist between various swap models, but usually the answers that competing models will come up with will be pretty close to one another.

²⁵A derivatives salesman once suggested to me that if I make a mistake pricing up a swap, I should be comfortable having the monetary value of that mistake subtracted from

that gives you an answer within a few tenths of a bp of a pricing model provides a great deal of comfort. Not every approximation provides that level of accuracy, however.

In order to motivate the approximation formulas that we will be using, suppose there is a bank that lets us borrow or lend money at the same rate of interest with semiannual compounding, denoted r_{sb} . If we were to lend \$1 to the bank today, at the end of one year our money will grow to

$$\$1 \times \left(1 + \frac{r_{sb}}{2}\right)^2. \quad (1.2)$$

And since we can borrow at the same rate, note that if we were to borrow \$1 from the bank today, we would owe the same amount at the end of a year. Suppose there is another bank that offers to borrow or lend at the same rate of interest with annual compounding, denoted r_{ab} . Suppose we invest \$1 in this bank today. At the end of one year, our money will grow to

$$\$1 \times (1 + r_{ab}). \quad (1.3)$$

Similarly, if we borrowed \$1 from this bank, we would owe this same amount at the end of the year. If the two quantities in Equations 1.2 and 1.3 were not the same, there would be arbitrage.²⁶ Thus, it must be that

$$\left(1 + \frac{r_{sb}}{2}\right)^2 = 1 + r_{ab}$$

or, rearranging terms, that

$$r_{ab} = \left[\left(1 + \frac{r_{sb}}{2}\right)^2 - 1\right]. \quad (1.4)$$

Although Equation 1.4 holds perfectly for compound rates of interest offered by banks as just described, it does not hold perfectly when comparing swap

my bonus at the end of the year. I told him I wasn't comfortable with anything being subtracted from my bonus. He said that was a very good answer.

²⁶As discussed in greater detail in Appendix A, an *arbitrage* is a trading strategy that enables us to lock in a riskless profit. To be a bit more formal, an arbitrage is a strategy in which we have nonnegative cash flows today and nonnegative cash flows in all states of nature in the future, with strictly positive cash flow either today or in at least one state of nature in the future. We will assume throughout that there are no arbitrage opportunities.

Suppose that the bank that quoted its interest rate on a semiannually compounded basis paid us more money at the end of the year. If this were the case, we could borrow \$1 at the other bank and invest it at the bank paying interest with semiannual compounding. At the end of the year, we would take the money we earned in the bank and have more than enough proceeds to pay back our loan. Thus, we have zero cash flows today and positive cash flows in the future, which would be an arbitrage. And if the quantity in Equation 1.3 were greater than the quantity in Equation 1.2, we could reverse the strategy, which would also be an arbitrage.

rates of different payment frequencies.²⁷ That being said, the underlying principle used to derive Equation 1.4 is at work when we consider changing the frequency of the fixed rate in a swap, and we can use this equation to approximate the relationship between annual bond and semi bond swap rates. To this end, for any given maturity, define s_{ab} and s_{sb} as the annual bond and semi bond swap rate, respectively. Then we have

$$s_{ab} \approx \left[\left(1 + \frac{s_{sb}}{2} \right)^2 - 1 \right]. \quad (1.5)$$

And what if we wanted to convert an annual bond swap rate to an annual money swap rate? Given an annual bond swap rate s_{ab} , a good approximation to the annual money swap rate, denoted s_{am} , is given by²⁸

$$s_{am} \approx \frac{360}{365} \times s_{ab}. \quad (1.6)$$

Example 1

Referring to Table 1.1, the offer side 30-year swap rate is 3.583% semi bond. Suppose we are interested in the offer side 30-year swap rate quoted on an annual bond basis. A good approximation for the 30-year swap rate quoted on an annual bond basis is found using Equation 1.5. We see that if the 30-year swap rate is 3.583% semi bond, then the 30-year swap rate quoted annual bond should be approximately

$$\left[\left(1 + \frac{.03583}{2} \right)^2 - 1 \right] = 3.615\%.$$

Notice that the annual bond swap rate is greater than the semi bond swap rate. The intuition for this result comes from the two banks that we considered that offer interest on either an annual or semiannual basis: the fixed rate receiver who receives annually does not have the opportunity to reinvest coupons every six months, and thus the annual rate must be higher to compensate him. $\square$

²⁷What would make you think that applying Equation 1.4 to swaps would be a valid thing to do? This relationship was formed by the concept of investing money in a bank and determining a restriction through an assumed lack of arbitrage. We can't even get off the ground with this type of reasoning for swaps. After all, in a swap there is no notion of investing any money today inasmuch as a plain vanilla swap has no upfront cash flows.

²⁸If a rate is accrued on a money basis, then it will accrue an extra five days (or six days in the case of a leap year) per annum. This formula adjusts the rate for those additional days. It may be more exact to let the denominator equal to 365.25, the average number of days in a year, but most people just live with using 365 in the denominator.

Example 2

Continuing with Example 1, if the 30-year swap rate is 3.615% annual bond, then a decent approximation to the swap rate quoted on an annual money basis can be found using Equation 1.6. In particular, the annual money rate should be approximately

$$3.615\% \times \frac{360}{365} = 3.565\%. \quad (1.7)$$

The annual money rate is lower than the annual bond rate, reflecting the fact that the fixed rate receiver in the annual money swap gets the benefit of accruing the fixed rate for an extra five or six days a year (depending on whether or not it's a leap year).

In order to see why this is, consider again a notional of \$100mm on the swap and assume that we receive fixed. Assuming the fixed side pays annual bond, the fixed cash flows that we will receive each year are equal to

$$\$100\text{mm} \times 3.615\% \times \frac{360}{360} = \$3,615,000.$$

Suppose instead that the swap pays annual money on the fixed side and further assume that there are 365 days in the year (i.e., it is not a leap year). Then if we could receive annual money at the same fixed rate of 3.615%, at the end of the year we would receive

$$\$100\text{mm} \times 3.615\% \times \frac{365}{360} = \$3,665,208.$$

Although we would strongly prefer then to receive the same fixed rate paid annual money (as doing so would mean we would receive \$3,665,208 per year versus only \$3,615,000), our counterparty would surely be opposed to this. We could ask ourselves the question, What fixed rate paid annual money would also provide \$3,615,000 at the end of each year, thereby making each counterparty indifferent to the change? This rate, denoted s_{am} , solves

$$\$100\text{mm} \times s_{am} \times \frac{365}{360} = \$3,615,000,$$

or $s_{am} = 3.565\%$, as per the approximation we obtained in Equation 1.7. $\square$

To summarize our combined findings from Example 1 and Example 2, if the 30-year swap rate is 3.583% semi bond, then a good approximation if the rate were instead quoted annual money is 3.565%, or about 1.8 bps lower. Referring to Table 1.1, we see that this approximation is two-tenths

of a basis point away from the annual money offer side rate of 3.563% shown on the composite page and would therefore be considered by many in the industry to be a decent approximation.

In the example we just considered, the semi bond swap rate of 3.583% was higher than the annual money equivalent of 3.563% as quoted on the composite page. Thus, the effect of changing the day count dominated the effect of changing the frequency. This need not always be the case. For example, if the semi bond swap rate is 7%, it can be shown using Equations 1.5 and 1.6 that the annual money swap rate is approximately 7.025%, and thus in this case the semi bond swap rate is actually below its annual money equivalent.

Think all of this rate conversion stuff is a waste of time? Then consider the following: if someone agrees to pay fixed annual money in a \$100mm 30-year swap but forgets to convert the rate as just shown and pays a rate of 3.583% instead of 3.563%, this mistake would cost around \$365,000 in present value terms, an amount we will be able to verify in Chapter 2.

Generalized Payment Frequency and Day Count Conversion Formulas

We can generalize the approximation formula in Equation 1.5 for arbitrary payment frequencies. If s_{xb} is a swap rate paid x times per year on a bond basis, and s_{yb} is the equivalent swap rate paid y times per year on a bond basis, then

$$s_{yb} \approx y \times \left[\left(1 + \frac{s_{xb}}{x} \right)^{x/y} - 1 \right]. \quad (1.8)$$

And given s_{yb} , a swap rate paid y times per year on a bond basis, we can approximate s_{ym} , the equivalent swap rate paid y times per year on a money basis as

$$s_{ym} \approx \frac{360}{365} \times s_{yb}. \quad (1.9)$$

Equation 1.8 should only be used for rates that are on a bond basis. If you need to convert back and forth between a bond and money basis, use Equation 1.9. Once again, it is very important to note that Equations 1.8 and 1.9 are only approximations and not exact formulas, as far as being employed to convert swap rates from one payment frequency to another or from one day count basis to another. For practical purposes, they provide decent approximations but are not a substitute for exact numerical answers from a pricing model.